

Exploring how dietary patterns shape the global obesity crisis

How our diets shape our destiny: A data driven journey into global obesity

Course Name & Code: Data Visualisation and Communication (**MATH2270**)

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Introduction

A Global Weight Gain Story

In the last few decades, our plates and waistlines have changed dramatically.

Once, traditional home-cooked meals nourished families. Today, processed snacks and fast food dominate diets across both developed and developing nations. With drive-thrus replacing dinner tables, the world is gaining weight fast.

Obesity is now a global epidemic, affecting over **650 million adults worldwide** (WHO, 2024). The culprits? A rapid rise in **high-calorie, nutrient-poor diets**, often rich in sugar, saturated fats, and refined carbohydrates.

As countries industrialize, fast food becomes more accessible and affordable gradually replacing traditional diets once rich in fibre, plant-based protein, and whole grains.

This dietary shift leads to an **energy imbalance**, causing excess body fat accumulation and elevating risks of **type 2 diabetes, heart disease, and cancer**.

How do changing intakes of sugar, fat, protein, and carbs shape the global obesity crisis?

This project explores that question using global data from **1990 to 2022**, investigating the correlations between **macronutrient intake** and **obesity rates** over time and geography.

Data Sources & Preprocessing

Where Our Data Comes From

To explore the nutrition-obesity link, we combined **three open global datasets**:

- **Obesity Rates (WHO GHO)**
> Age-standardized % of obese adults ($\text{BMI} \geq 30$), 1990–2022
- **Diet Composition (FAO)**
> Per capita daily kilocalories from food groups like sugar, fats, oils, etc.
- **Macronutrient Supply (Our World in Data)**
> Average kcal/day from carbohydrates, fat, and animal vs. plant protein

Preprocessing Steps

- Merged datasets by Country and Year
- Selected variables: **Obesity Rate, Sugar, Fat, Carbs, Animal & Plant Protein**
- Filtered for complete cases (no missing kcal values)
- Final dataset: **5,600+ country-year observations (1990–2022)**

This cleaned dataset is the foundation of our interactive visual analysis.

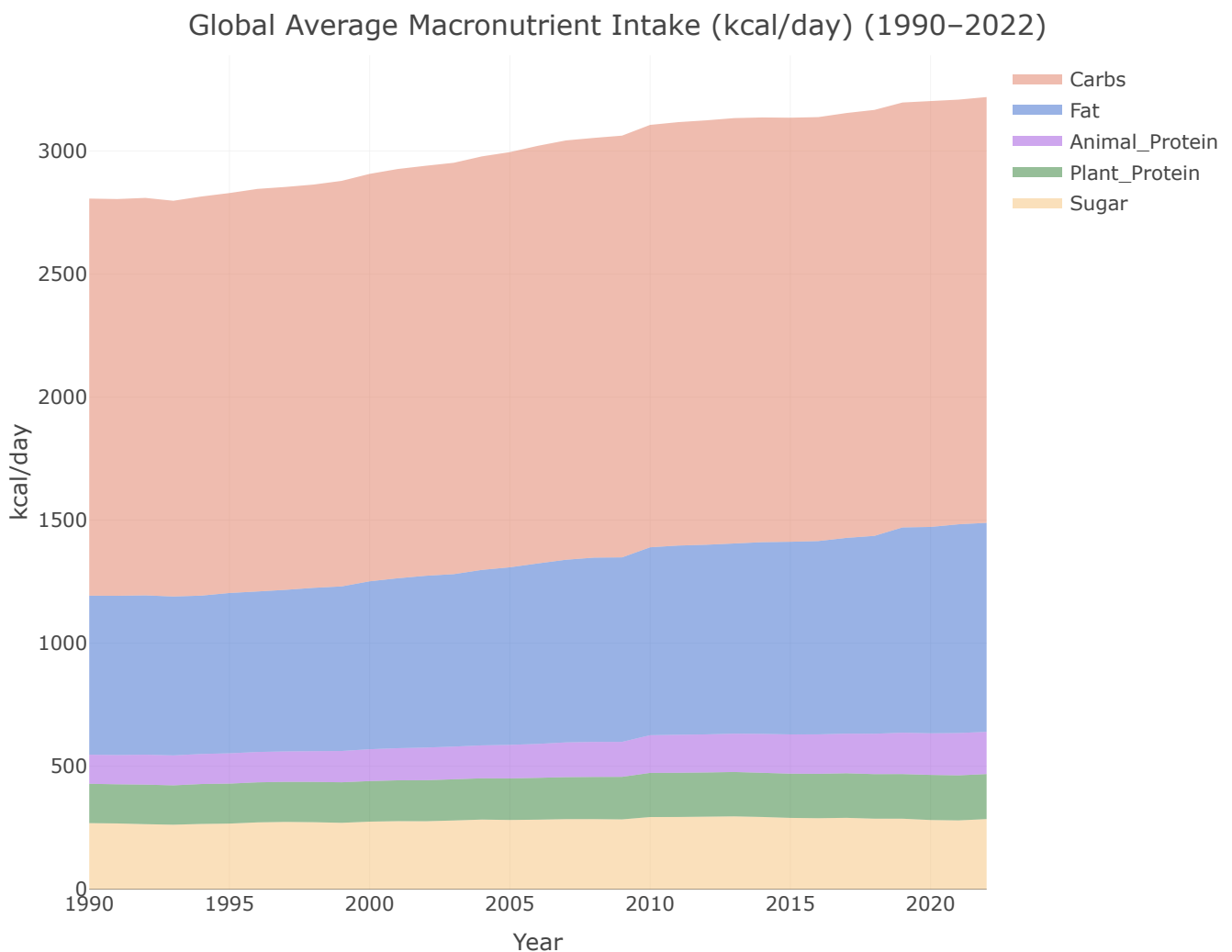
Global Macronutrient Intake Trends (1990–2022)

Are We Eating Differently Over Time?

This stacked area chart visualizes the average global intake (in kcal/day) of five macronutrients from 1990 to 2022:

Key Observations:

- **Carbohydrates** remain the largest energy source, though their share has slightly declined.
- **Fat intake** has steadily increased, reflecting dietary shifts toward processed and high-fat foods.
- **Sugar consumption** rose modestly, especially post-2000.
- **Animal protein** has gradually overtaken **plant protein** in global diets, especially in industrialized nations.



Implication:

These macronutrient shifts reflect a **nutrition transition** from traditional diets to modern, energy-dense foods that aligns with rising obesity rates.

Let's see how this connects with the global obesity curve.

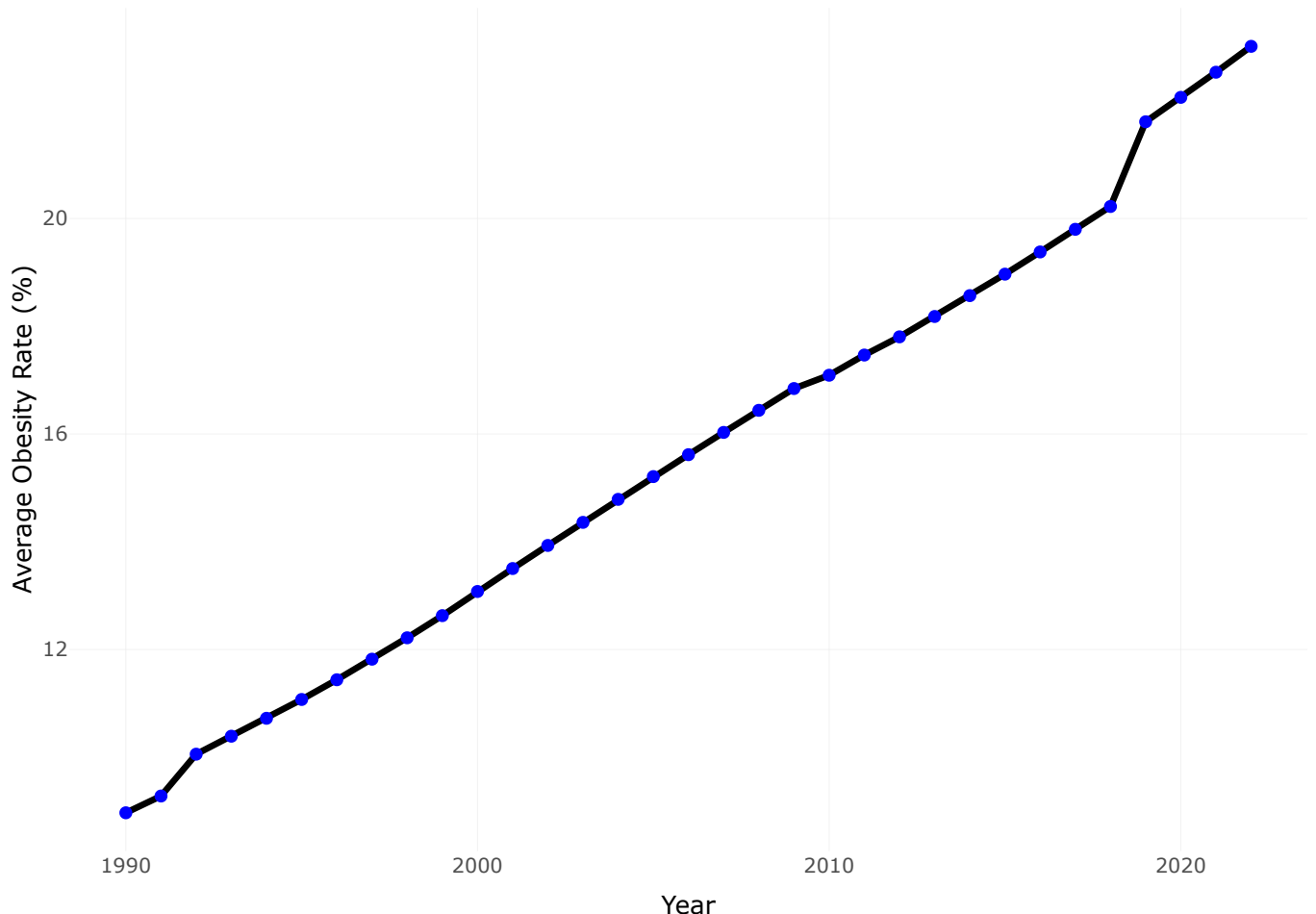
Global Obesity and Nutrition Trends (1990–2022)

How Has Global Nutrition Shifted?

Between **1990** and **2022**, the world witnessed a steady **rise in obesity rates** alongside noticeable dietary shifts:

- 🍬 **Sugar intake** increased, particularly in urbanized and high-income regions.
- 🍔 **Fat consumption** (especially from animal sources) rose significantly.
- 🥩 **Animal protein** replaced many traditional plant-based staples.
- 📉 **Carbohydrate intake** plateaued or declined slightly in many regions.

Global Average Obesity Rate Over Time



What the Data Shows

- The **global obesity rate** has more than **doubled**, climbing from ~10% in 1990 to **over 23%** by 2022.
- The trend aligns with increased **energy-dense, nutrient-poor diets**.

- The sharpest growth occurred **post-2000**, coinciding with fast food proliferation and lifestyle changes.

This global rise signals a deeper nutritional shift.

Up next: What **macronutrients** are most strongly linked to this pattern?

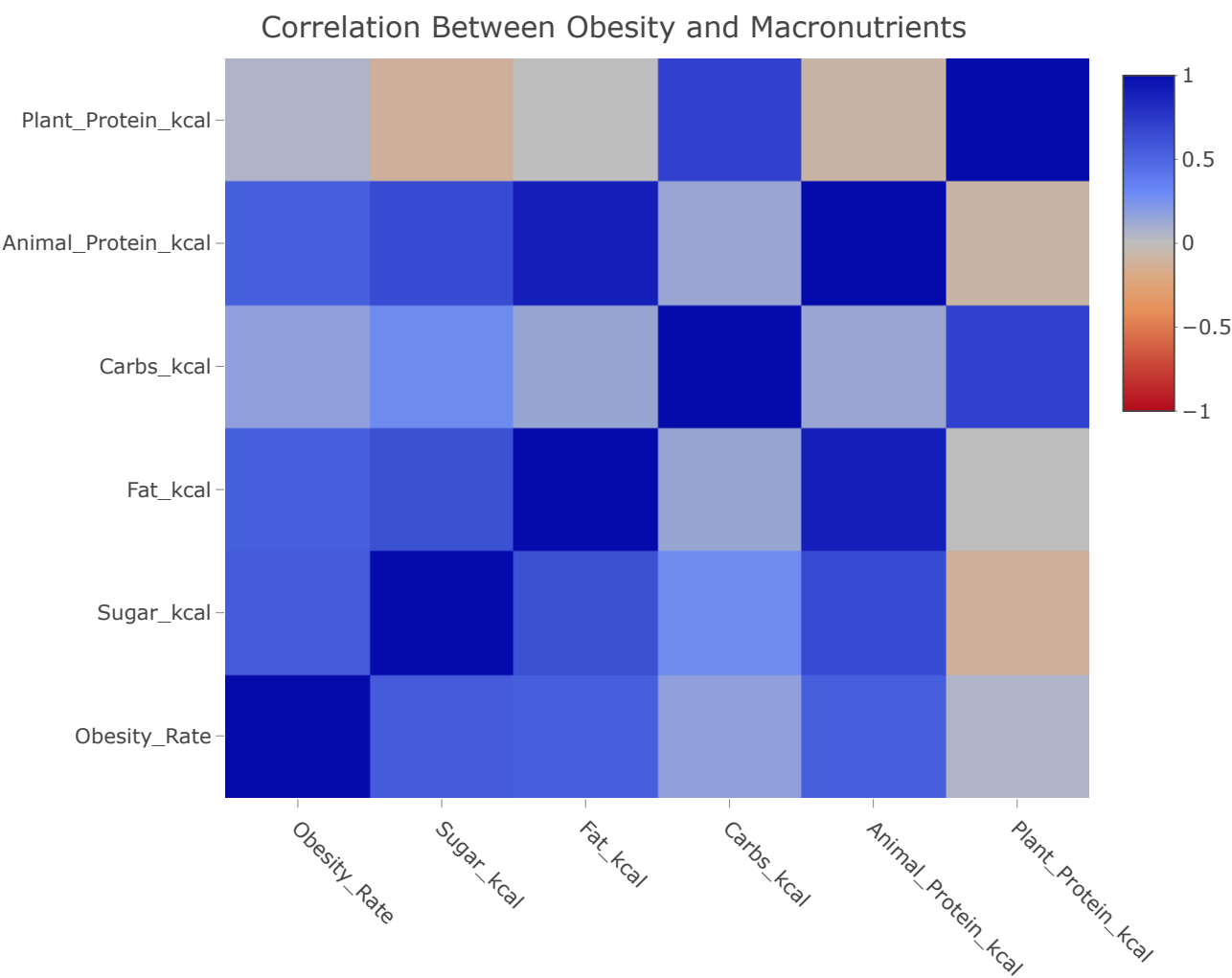
What Drives Obesity? Macronutrient Correlations

Identifying Key Dietary Drivers

To investigate potential contributors to the global rise in obesity, we examined correlations between macronutrient intake and obesity rates from 1990 to 2022.

Key Insights:

- **Sugar** and **fat intake** are strongly and positively correlated with obesity rates.
- **Animal protein** shows moderate positive correlation.
- **Carbohydrates** have a weak or slightly negative correlation.
- **Plant protein** exhibits a weak negative association.



Interpretation:

Diets high in sugar and fat appear to be key contributors to rising obesity rates, while plant-based diets may offer a protective effect.

This evidence highlights the nutritional transition driving global obesity trends.

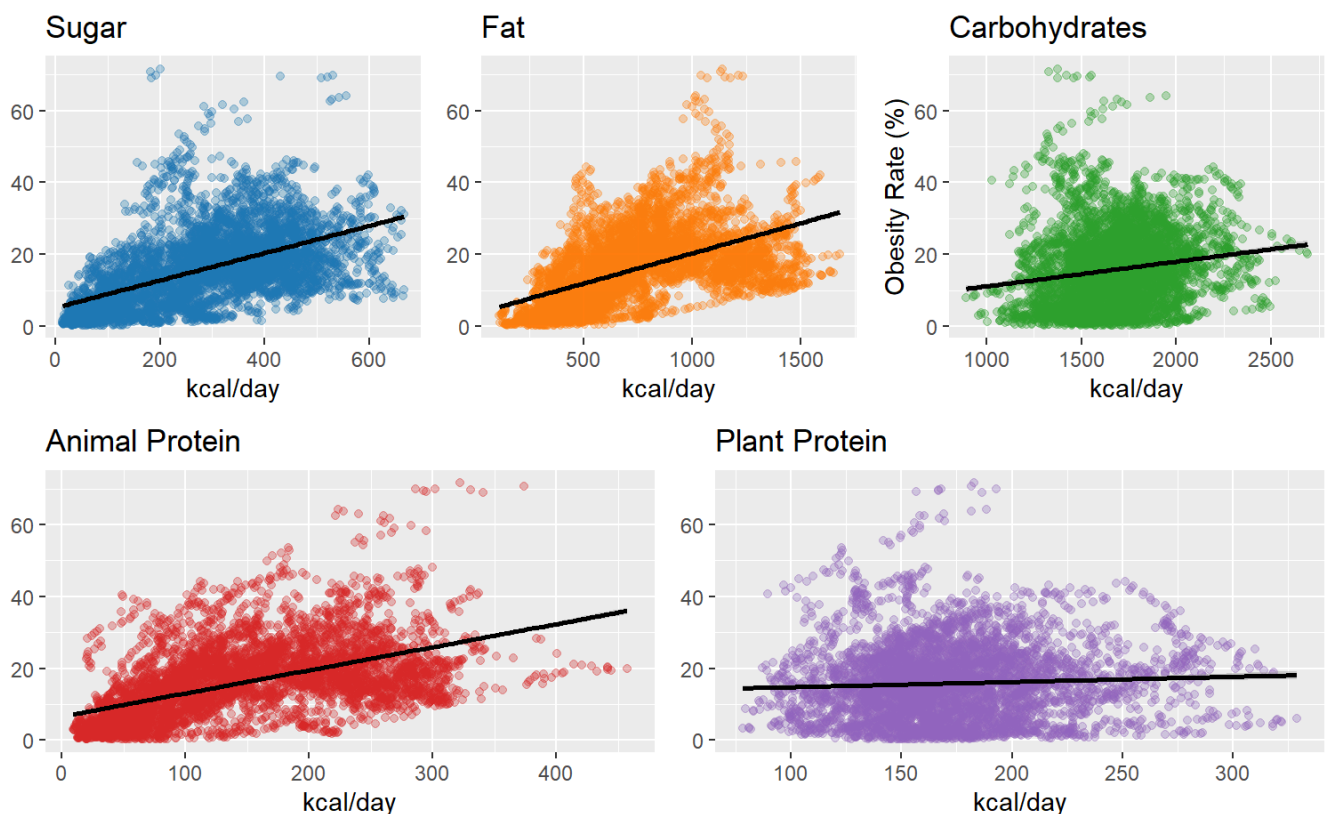
Next: We explore how each macronutrient individually relates to obesity rates.

Obesity vs. Macronutrient Intake

How Does Each Nutrient Influence Obesity?

This section explores how specific macronutrients correlate with obesity rates based on global data (1990–2022):

- **Sugar:** Strong positive correlation countries with higher sugar intake generally have higher obesity rates.
- **Fat:** Also positively associated with obesity, though less consistent than sugar.
- **Carbohydrates:** Weak or no clear association some high-carb diets correspond with lower obesity rates.
- **Protein:**
 - *Animal protein shows a mild positive trend.*
 - *Plant protein tends to have a neutral or slightly protective effect.*



Summary: What Drives Obesity?

- Obesity has more than doubled globally since 1990.
- Sugar and fat intake show strong positive relationships with obesity.

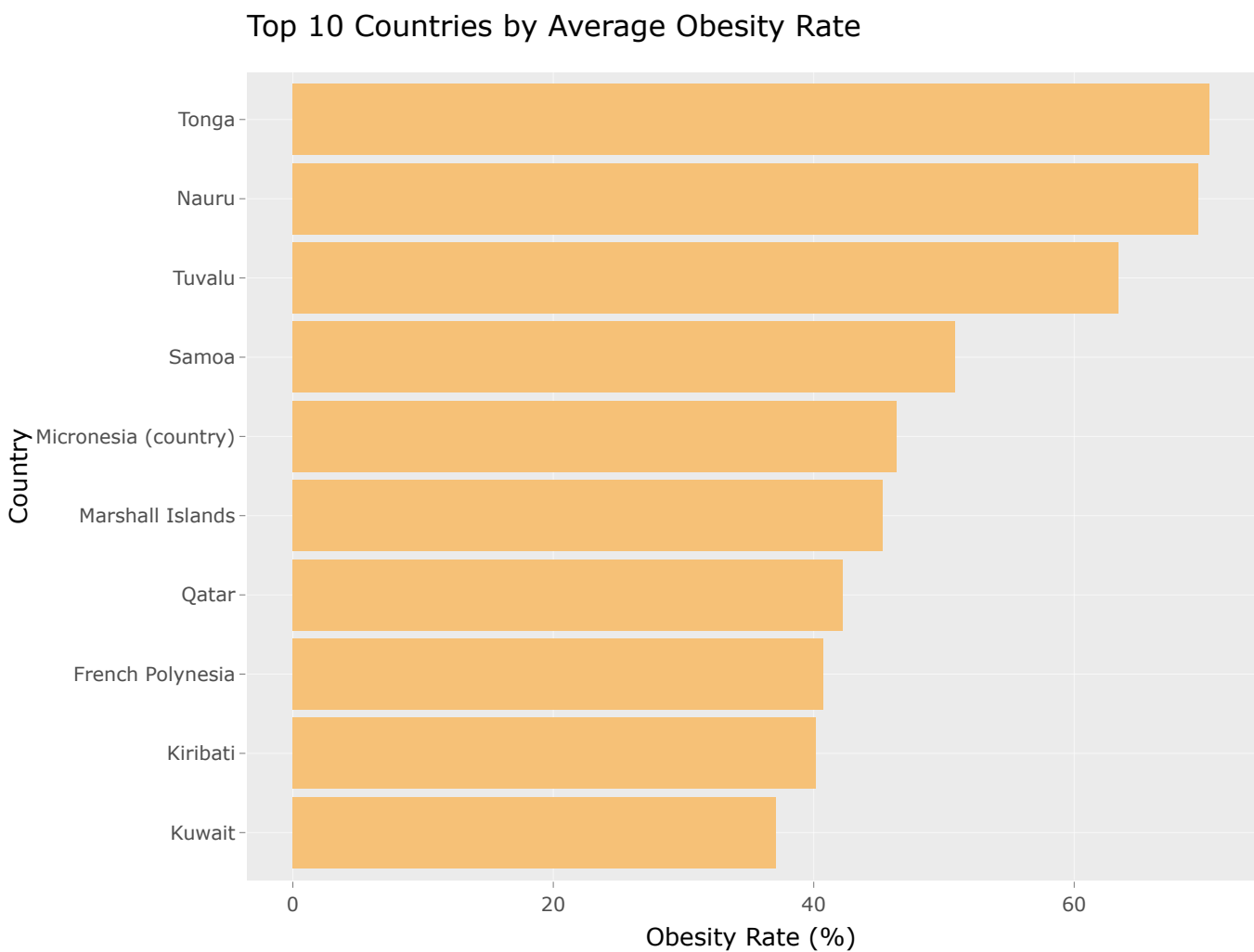
- Plant-based diets appear to offer modest protective effects.
- The transition to energy-dense, processed foods is a major global concern.

These patterns offer a big-picture view, but which regions are most affected?

Country Focus:Top 10 by Average Obesity Rate (1990–2022)

- Small island nations in the **Pacific region** dominate the list.
- These countries have undergone rapid **dietary westernization**, often replacing traditional, nutrient-dense diets with imported processed foods.
- **Tonga, Nauru, and Tuvalu** consistently show average obesity rates above **60%**.
- High prevalence is linked to affordability of processed imports and limited access to fresh, local foods.

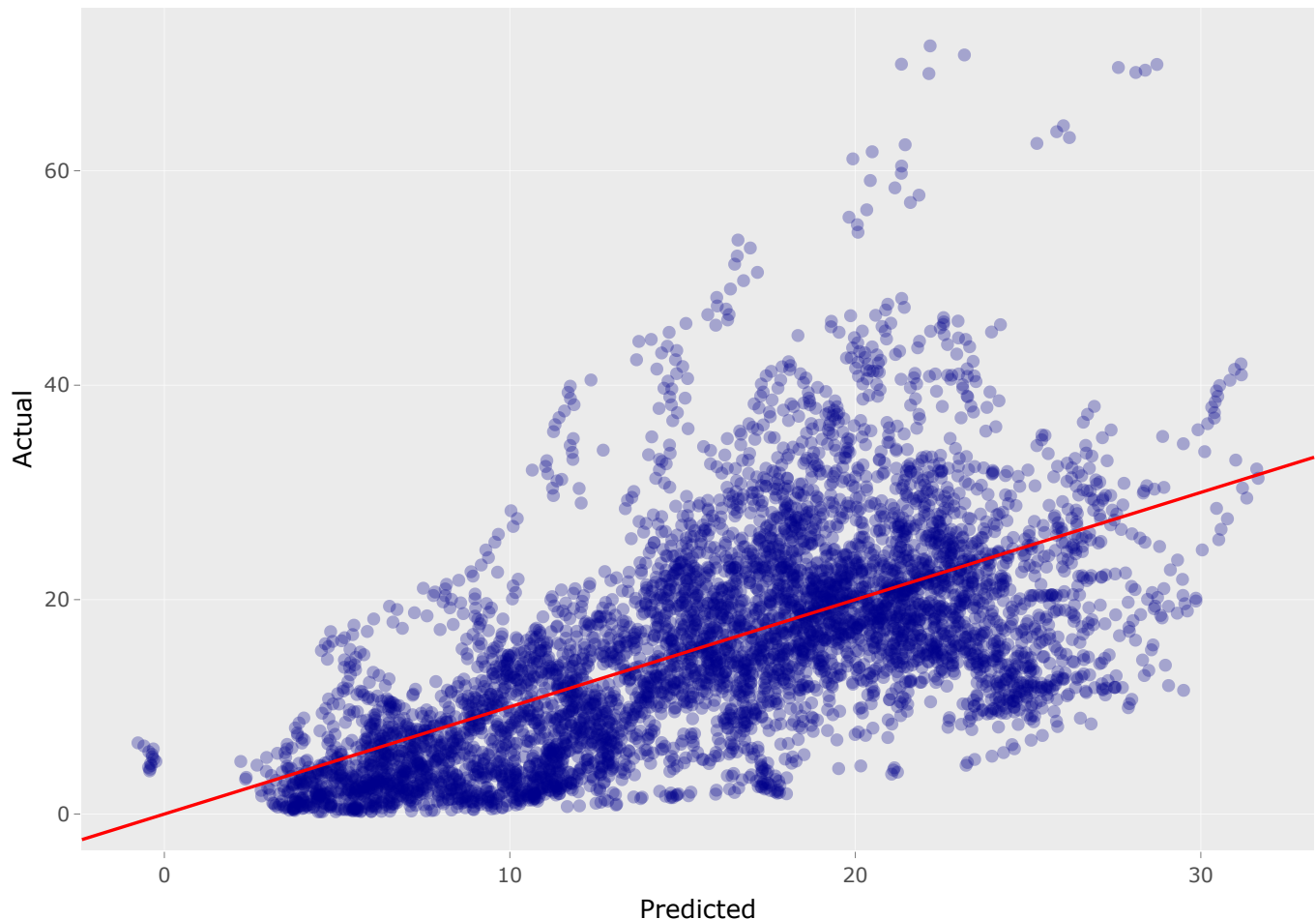
*Global solutions must consider **localized dietary shifts** and **cultural influences**.*



Can We Predict Obesity From Diet?

- A **linear regression** shows that sugar intake alone explains ~**29%** of the variance in obesity.
- A **multiple regression model** using sugar, fat, carbs, and proteins improves explanatory power to ~**37% ($R^2 = 0.37$)**.

Predicted vs Actual Obesity Rate



What the Prediction Plot Shows:

- The red line represents perfect prediction (Actual = Predicted).
- The scatter shows some variability indicating **other lifestyle, economic, and genetic factors** also play a role.
- Still, **macronutrient intake** has a measurable and statistically significant effect on obesity trends.

Conclusion

Over the past three decades, global obesity rates have **more than doubled**, rising alongside significant shifts in dietary patterns. This analysis uncovered clear patterns:

- **High intake** of sugar, fat, and animal protein is **strongly linked** to higher obesity rates.
- Nations most affected including many **Pacific islands** and **resource-rich countries** are experiencing rapid **nutrition and lifestyle transitions**.
- In contrast, **plant-based diets** and **moderate sugar consumption** appear to offer **protective benefits**. Together, these findings paint a powerful picture: **Obesity is not just a personal health issue it's a global nutritional consequence.**

Effective policy, education, and access to healthier foods are essential to reverse this rising tide.

Reference

[1] Ritchie, H., & Roser, M. (2024). Obesity prevalence among adults (WHO). Our World in Data. <https://ourworldindata.org/grapher/obesity-prevalence-adults-who-gho>

[2] Ritchie, H., & Roser, M. (2024). Daily caloric supply from carbohydrates, protein and fat. Our World in Data. <https://ourworldindata.org/grapher/daily-caloric-supply-derived-from-carbohydrates-protein-and-fat>

[3] Ritchie, H., & Roser, M. (2024). Dietary composition by country. Our World in Data. <https://ourworldindata.org/grapher/dietary-composition-by-country>